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ORIGINAL ARTICLE

"ChatGPT can make mistakes. Check important info." Epistemic beliefs and metacognitive accuracy in students' integration of ChatGPT content into academic writing

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Recent studies have conceptualized ChatGPT as an epistemic authority; however, no research has yet examined how epistemic beliefs and metacognitive accuracy affect students' actual use of ChatGPT-generated content, which often contains factual inaccuracies. Therefore, the present experimental study aimed to examine how university students integrate correct and incorrect information from expert-written and ChatGPT-generated articles when writing independently ($N = 49$) or with ChatGPT assistance ($N = 49$). Students working with ChatGPT-4o integrated more correct information from both expert-written ($d = 0.64$) and ChatGPT-generated articles ($d = 0.95$), but ChatGPT-assisted writing did not affect the amount of incorrect information sourced from the ChatGPT-generated article. Regardless of the condition, hierarchical regressions revealed that lower metacognitive bias was moderately associated with increased inclusion of correct information from the expert-written article ($R^2 = 12\%$). Conversely, a higher metacognitive bias ($R^2 = 10\%$) and epistemic beliefs ($R^2 = 12\%$) were moderately related to the inclusion of incorrect information from ChatGPT-generated articles. These findings suggest that while ChatGPT assistance enhances the integration of correct human- and AI-generated content, metacognitive skills remain essential to mitigate the risks of incorporating incorrect AI-generated information.

KEYWORDS

ChatGPT, epistemic cognition, generative artificial intelligence, metacognition, metacognitive accuracy, multiple text reading

Practitioner notes

What is already known about this topic

- Generative AI tools, such as ChatGPT, are increasingly regarded as epistemic authorities due to their authoritative tone and human-like interaction.
- ChatGPT has demonstrated utility in providing correct information and improving productivity in educational and professional contexts, but it is also prone to inaccuracies, hallucinations and misleading content.
- Students' epistemic beliefs and metacognitive skills predict their ability to critically evaluate and integrate conflicting information from multiple resources, particularly when searching for information on the Internet.

What this paper adds

- This study experimentally examines how students integrate correct and incorrect information from expert-written and ChatGPT-generated articles when writing independently or with ChatGPT's assistance.
- The findings show that ChatGPT assistance improves the inclusion of correct information but does not significantly reduce or increase the inclusion of incorrect ChatGPT-generated content.
- Metacognitive accuracy and epistemic beliefs are key factors in mitigating the inclusion of incorrect information, regardless of whether students work independently or with ChatGPT.

Implications for practice and/or policy

- Generative AI tools can outperform human experts in specific scenarios, requiring little to no evaluation. However, in situations where these tools generate misleading or incorrect content, the application of metacognitive skills and epistemic beliefs becomes essential to discern reliable information and avoid the integration of errors.
- Educational interventions should include activities requiring justification of knowledge, evaluation of resources and reflection upon human-generated and AI-generated texts to enhance students' ability to discern accurate from inaccurate information.
- Interventions focused on metacognitive accuracy and epistemic awareness can empower individuals to critically evaluate and differentiate between reliable and erroneous information, enhancing their recognition of misinformation.

INTRODUCTION

Working with generative AI tools, such as ChatGPT, increases productivity in complex problem-solving tasks in both educational and professional settings (Noy & Zhang, 2023; Sun & Zhou, 2024; Urban et al., 2024). These tools excel in standardised knowledge tests (Sabri et al., 2025), verbal ability tasks (Hickman et al., 2024) and creativity tasks (Haase & Hanel, 2023), outperforming most human participants. With their proven utility, authoritative tone in generated text and chat-based interfaces that mimic human interaction, users may come to rely on generative AI tools as ultimate epistemic authorities (Cooper, 2023). Although the term 'epistemic authority' faced substantial criticism—since these tools lack genuine reasoning which is foundational for traditional conceptualisations of epistemic experts (Ferrario et al., 2024; Monteith et al., 2024)—trust or distrust in these tools proved to have significant practical consequences (Goh et al., 2024; Messeri & Crockett, 2024).

Generative AI tools produce outputs that often appear highly credible, but are simultaneously prone to factual errors, fabricated sources and misleading content (Chelli et al., 2024; Ferrario et al., 2024). Trust in these tools can lead to overreliance, discouraging users from exploring alternative perspectives and reducing the variability of ideas, which results in more homogeneous outputs (Ding et al., 2023; Doshi & Hauser, 2024). In a scientific context, trust in generative AI can lead to the emergence of 'monocultures of knowledge', when AI-driven findings appear diverse, but they in fact restrict exploration to topics aligned with AI's capabilities (Messeri & Crockett, 2024). These findings, however, contrast with research in practical domains. Goh et al. (2024) found that medical practitioners using generative AI tools performed significantly worse in diagnosing clinical cases compared to AI tools themselves. Despite generative AI providing correct answers, the distrust led practitioners to reject them. The ambiguous nature of generative AI tools is further emphasised by Huang et al. (2024), who examined ChatGPT's responses to myths about Alzheimer's disease. Although ChatGPT consistently identified myths as false, some responses failed to highlight critical distinctions within the topic, potentially leading to misinterpretations.

These examples underscore the dual nature of generative AI tools: They have the potential to generate accurate and reliable knowledge with high utility value but can also produce misleading or partially incorrect information. The challenge, therefore, lies in determining when it is appropriate to rely on AI-generated knowledge and when caution or further verification is necessary. This is heightened by the fact that even experts, such as professional linguists and teachers, are unable to effectively distinguish between human-written and AI-generated text (Casal & Kessler, 2023; Fleckenstein et al., 2024). Furthermore, individuals unfamiliar with a topic tend to trust well-written but incorrect information, especially when it aligns with their expectations or preconceptions (Monteith et al., 2024). Finally, exposure to authoritative sources can influence how individuals integrate conflicting views, often reducing critical and evaluative thinking (Ku et al., 2014).

As outlined above, the challenge associated with AI-generated knowledge involves the interplay of two key constructs. First, epistemic beliefs reflect individuals' perspectives on the nature and process of knowing, which can be organised into four dimensions. The *certainty of knowledge* ranges from the belief that knowledge is fixed and absolute to the view that it is tentative and evolving. The *structure of knowledge* captures whether knowledge is understood as discrete, isolated facts or as interconnected, complex systems. The *source of knowledge* reflects whether individuals view knowledge as externally transmitted, residing in authoritative sources, or as actively constructed through personal reasoning and interaction. Finally, the *justification for knowing* refers to the need to evaluate knowledge, ranging from reliance on authority to critical assessments using evidence and comparisons across multiple sources (Bråten et al., 2005). Second, metacognition, which encompasses knowledge about cognition and the regulation of cognitive processes (Hacker, 1998), influences how individuals reflect on and justify their knowledge. This situated reflective activity, known as epistemic metacognition, involves evaluating the simplicity, certainty, source and justification of knowledge, which in turn shapes epistemic beliefs (Mason et al., 2010).

The present study, therefore, aims to examine how epistemic beliefs and metacognitive accuracy predict the integration of correct and incorrect knowledge from expert-written and ChatGPT-generated articles in university student writing under two conditions: working independently and working with ChatGPT-4o.

Epistemic beliefs

Epistemic beliefs refer to the views of individuals about the nature, justification and truth of knowledge. The rise of the Internet as a primary source of information has shifted the responsibility for evaluating the credibility and relevance of knowledge from publishers to readers, making epistemic beliefs more critical than ever (Strømsø & Kammerer, 2016). Strømsø and Bråten (2010) adapted the Internet-Specific Epistemological Questionnaire (ISEQ, Bråten et al., 2005) to measure epistemic beliefs about the certainty, structure and justification of Internet-based knowledge. In their eye-tracking study, Kammerer et al. (2013) found that individuals who believed that Internet-based knowledge is certain, reliable and well-structured were less likely to critically examine the credibility of the sources they encountered. In other words, they tended to accept the information at face value without questioning its validity. Furthermore, these beliefs were associated with individuals who displayed overconfidence in their decisions based on the information they found, even when they had not thoroughly evaluated its accuracy or trustworthiness. It was previously found that individuals relying heavily on Internet-based sources are more likely to view knowledge as certain, unchanging and handed down by an authority (Ståhl et al., 2021). Over-reliance on Internet-based sources and a lack of critical evaluation pose risks for the quality of knowledge construction (Ståhl et al., 2021; Strømsø & Kammerer, 2016), epistemic beliefs in omniscient authority being associated with weaker argumentation and less engagement in counter-reasoning (Ku et al., 2014). This behaviour highlights how certain epistemic beliefs can reduce critical engagement with information and lead to potentially flawed conclusions. This aligns with the findings of Richter and Schmid (2010), who observed that when people believed that knowledge is uncertain, they were more likely to use strategies to check for consistency in the information they encountered.

Epistemic beliefs guide self-regulated learning by shaping goal-setting, knowledge evaluation, strategy selection and metacognitive monitoring and control (Bromme et al., 2010; Muis, 2007). Epistemic beliefs, therefore, help learners navigate complex tasks, adapt to diverse learning environments and critically engage with multiple knowledge sources (Muis & Singh, 2018). The meta-analysis by Greene et al. (2018) found that epistemic beliefs positively correlate with academic achievement across educational stages, from elementary school to graduate studies, though the overall effect size is small ($r = 0.16$). Despite the importance of epistemic beliefs in working with multiple information sources, there is currently no clear conceptualisation of epistemic beliefs about AI-generated knowledge. This study aims to address this gap by exploring the role of domain-specific beliefs about ChatGPT-generated knowledge and task-specific beliefs about both expert-written and ChatGPT-generated articles.

Metacognition

Epistemic metacognition involves metacognitive knowledge and metacognitive monitoring and regulation specifically oriented towards knowledge and knowing (Barzilai & Zohar, 2014; Hofer, 2004). It involves assessing the coherence and credibility of evidence and justification of information. By engaging in epistemic metacognition, individuals can critically work with conflicting information, applying their epistemic beliefs to guide their reasoning (Barzilai & Zohar, 2014; Hofer, 2004; Maier & Richter, 2014).

Bråten et al. (2011) proposed that students with more sophisticated epistemic beliefs are more likely to activate and utilise metacognitive strategies, such as planning, monitoring, regulation and evaluation, when encountering conflicting information. Research by Chiu et al. (2013) corroborated this proposition when students with sophisticated beliefs about justification, such as validating information through multiple sources, were better at initiating self-regulatory processes. However, effectively applying metacognitive strategies required recipients to actively monitor the plausibility of information and regulate their cognitive and behavioural processes accordingly (Maier & Richter, 2013). For instance, students who critically evaluated Internet-based knowledge demonstrated a higher likelihood of engaging in goal setting and planning (Chiu et al., 2013).

Individuals with sophisticated epistemic beliefs (ie, recognising knowledge as interconnected, evolving, and emerging from evaluation of evidence; Greene et al., 2018) approached conflicting information with a more analytical approach (Bråten & Strømsø, 2006). Even eighth graders who viewed scientific knowledge as complex and evolving justified online information through comparisons of multiple sources and engaged in higher-level reflections on the uncertainty of knowledge (Mason et al., 2010). Similarly, among ninth-grade students, stronger justification beliefs were associated with higher integration performance when resolving conflicting information (Barzilai & Ka'adan, 2017).

People are generally unaware of their susceptibility to inaccurate information. Those with higher confidence in their ability to resist inaccuracies often made more errors, revealing a lack of metacognitive calibration (Salovich & Rapp, 2021). Calibration, often operationalised as metacognitive bias, refers to alignment between judgements and actual performance, with low bias indicating underconfidence and high bias indicating overconfidence about one's performance (Schraw, 2009). Students with less developed epistemic beliefs are more prone to overconfidence (Kammerer et al., 2013).

In the context of generative AI use, Urban et al. (2024) found that metacognitive bias in university students solving non-routine problem-solving tasks was linked to their perceived usefulness of ChatGPT rather than their actual performance. In other words, students who used ChatGPT as an aid to solve a problem were unable to calibrate their judgments based on their performance. Instead, they relied on the perceived utility of the tool as a cue for their judgments. The present study therefore aims to investigate the role of metacognitive bias in the integration of information from expert-written and ChatGPT-generated articles.

Present study

The present study aims to examine how participants integrate information from two distinct sources—a medical expert-authored text that contains only correct information and a ChatGPT-generated text that contains both correct and incorrect information—when writing a brief article about the effects of vitamin C on the common cold under two conditions: independently and with the assistance of ChatGPT. This design was chosen to reflect a realistic scenario in which learners must critically engage with information from sources of varying reliability, as highlighted in multiple-text reading research (eg, Bråten et al., 2020;

Britt & Rouet, 2020; Strømsø et al., 2013). By simulating such complexity, the study investigates how epistemic beliefs influence students' evaluation and integration of reliable and less reliable knowledge sources.

The study builds on previous research showing that ChatGPT can generate a large number of ideas (Doshi & Hauser, 2024; Haase & Hanel, 2023) while producing a mix of largely correct (Sabri et al., 2025) and partially incorrect information (Chelli et al., 2024; Ferrario et al., 2024; Huang et al., 2024). In other words, these models tend to facilitate the inclusion of more content overall—both correct and incorrect. Based on these findings, the present study hypothesises **H1**: The student articles written with the assistance of ChatGPT will

(H1a) contain more correct information sourced from the medical article,
(H1b) contain more correct information sourced from the ChatGPT-generated article and
(H1c) contain more incorrect information sourced from the ChatGPT-generated article.

Furthermore, the study examines the role of metacognitive bias in the integration of information from expert-written and ChatGPT-generated articles. Previous studies suggested that human users often fail to recognise AI-generated text and are generally overconfident in their ability to do so (Casal & Kessler, 2023; Fleckenstein et al., 2024). Additionally, research has shown that higher metacognitive bias is associated with a greater likelihood of making errors when dealing with inaccuracies in texts (Salovich & Rapp, 2021), difficulties in reasoning about controversies (Lang et al., 2021), and greater resistance to correcting misinformed beliefs (Rapp & Withall, 2024). To assess metacognitive bias, the present study employed the Cognitive Reflection Test (CRT; Frederick, 2005), a measure of reflective reasoning and the ability to override intuitive, automatic responses. The CRT has been found to correlate with tasks involving deliberative reasoning, such as belief bias and denominator neglect (Toplak et al., 2013), which align with the cognitive demands of integrating conflicting information from reliable and less reliable sources. Based on these findings, the present study investigates **RQ1**: How does metacognitive bias predict the inclusion of correct information from expert-written articles and correct and incorrect information from ChatGPT-generated articles?

Finally, the study investigates the role of epistemic beliefs in the integration of information from expert-written and ChatGPT-generated articles. Richter and Schmid (2010) found that participants who viewed knowledge as uncertain were more likely to employ consistency-checking strategies. In contrast, Kammerer et al. (2013) observed that individuals with stronger beliefs about the certainty and reliability of Internet-based knowledge were less likely to critically evaluate the credibility of sources and displayed overconfidence in their ability to work with information. Based on these findings, the present study asks **RQ2**: How do epistemic beliefs predict the inclusion of correct information from expert-written articles and correct and incorrect information from ChatGPT-generated articles?

METHODS

Experimental procedure

Ethical approval for the study was obtained from the Ethical Committee of the first author's institution, following the ethical principles outlined by the APA.

University students were invited to participate via email, with the study described under the general title ‘Generative AI in education’. Participants were incentivised with a small portion of course credit for their involvement.

Students who volunteered were randomly assigned to either the control (independent writing) or the experimental group (ChatGPT-assisted writing) prior to the start of the study.

The sessions were conducted in a quiet laboratory with a maximum of five participants per session. Control and experimental groups were tested in separate sittings. Each participant worked on an individual computer in a private workspace, ensuring no interaction or visibility of other participants' screens. The computers were identical in hardware and software configurations.

Informed consent was administered electronically before the tasks commenced. Participants in both groups completed the Cognitive Reflection Test (Frederick, 2005) with accompanying metacognitive judgments. Then they completed the ChatGPT-specific epistemic beliefs questionnaire (ChatGPT-EBQ; see Appendix A). Finally, they read two texts on the effect of vitamin C on the common cold. Both texts were approximately 350 words long. The first text (Flesch Reading Ease Score = 46.9) was authored by a medical expert. It was based on Hemilä and Chalker's (2013) meta-analysis and contained only correct information (the maximum points for correct expert information was 11). The second text (Flesch Reading Ease Score = 46.3) was an unaltered outcome generated by ChatGPT-3.5 that included both correct (the maximum points for correct ChatGPT-generated information was 7) and incorrect information (the maximum points for incorrect ChatGPT-generated information was 4). Each text clearly identified its author, ensuring that participants were aware of its origin (medical expert vs. ChatGPT). After reading, participants judged text-specific epistemic beliefs by items inspired by Mason et al. (2010).

The participants were then instructed to write a brief article (maximum two paragraphs) for student newspapers about 'impact of vitamin C on a common cold'. In the control condition, the students wrote the article independently, while in the experimental condition, the students used ChatGPT-4o to assist in the writing process. To ensure ecological validity, the ChatGPT environment was neither fine-tuned nor artificially modified; participants interacted with the standard, unaltered ChatGPT interface to mirror real-world usage scenarios. No constraints were placed on participants' interactions with ChatGPT. Participants in the experimental condition received a brief description of basic interaction strategies with ChatGPT: (1) they were informed that ChatGPT responds to prompts, (2) effective prompts require detailed elaboration to generate accurate responses, and (3) users can refine responses by providing additional context or requesting a regeneration. Participants in both conditions had constant access to both source texts and there was no time limit on the task.

The average time to complete the tasks was $M = 43.7$ minutes ($SD = 8.3$) in the control group and $M = 46.6$ minutes ($SD = 7.9$) in the experimental group. All participants passed two attention checks and completed the required tasks; thus, no exclusions were made.

Participants

For the between-group comparison, the sample size was determined a priori using G*Power 3.1.9.6, with $\alpha = 0.05$, $\beta = 0.80$, and an expected effect size of Cohen's $d = 0.62$. This effect size was estimated based on average between-group differences reported by Urban et al. (2024). The calculation indicated that a minimum of 42 participants per group would be required.

In the present study, the control group included 49 university students and the experimental group also consisted of 49 participants. Detailed characteristics of the participants are presented in Table 1. As can be seen, there were no significant differences between the groups in terms of gender, age, field of study, academic level, prior knowledge about vitamin C or prior experience with ChatGPT. The sample was racially and ethnically homogenous.

TABLE 1 Detailed information about participants.

	Control ($N = 49$)	Experimental ($N = 49$)	Comparison
Gender			
Female	39	35	$\chi^2(2) = 1.61$, $p = 0.45$
Male	10	13	
Other	0	1	
Age	$M_{age} = 21.80$, $SD = 3.25$	$M_{age} = 22.31$, $SD = 4.20$	$t(96) = 0.68$, $p = 0.50$
Field of study			
Social sciences and humanities	39	39	$\chi^2(2) = 0.22$, $p = 0.90$
Life sciences	6	7	
Technical subjects	4	3	
Academic level			
BA	41	37	$\chi^2(1) = 1.01$, $p = 0.32$
MA	8	12	
Vitamin C prior knowledge	$M = 1.67$, $SD = 0.90$	$M = 1.69$, $SD = 0.78$	$t(96) = 0.08$, $p = 0.94$
ChatGPT prior experience	$M = 4.98$, $SD = 1.40$	$M = 4.80$, $SD = 1.38$	$t(96) = 0.58$, $p = 0.56$

Measures

Metacognitive accuracy (bias index)

A decontextualised metacognitive bias was calculated as the difference between metacognitive judgements and actual performance on a separate experimental task. This approach was used to avoid collinearity problems, allowing metacognitive bias to serve as a predictor in the analyses (Fleming et al., 2012; Fleming & Lau, 2014; Schraw, 2009; Urban & Urban, 2023). To obtain the performance level, participants completed the Cognitive Reflection Test (Frederick, 2005), which includes three items. For each item, the participants provided a metacognitive judgement by rating their confidence in their performance on a scale from 0 (absolutely unsure) to 100 (absolutely sure). The bias index was calculated by averaging the differences between the participants' confidence ratings and their actual performance on each item. The resulting score was standardised to range from −1 to 1. A bias index close to −1 indicates underconfidence, where participants completely underestimate their performance. A score near 0 reflects accurate monitoring. A bias index close to 1 indicates overconfidence, where participants completely overestimate their performance.

Epistemic beliefs about ChatGPT

To assess decontextualised epistemic beliefs about ChatGPT-generated knowledge, the present study adapted the three-factor Internet-Specific Epistemological Questionnaire developed by Strømsø and Bråten (2010). The three factors included Certainty and Source of Knowledge ($\omega = 0.77$), which assessed participants' beliefs about the completeness and reliability of ChatGPT as a source of knowledge; Structure of Knowledge ($\omega = 0.74$), which measured beliefs about the complexity of the knowledge generated by ChatGPT; and Justification for Knowing ($\omega = 0.80$), which evaluated participants' approaches to verify the ChatGPT-generated knowledge. Participants rated each item on a Likert scale ranging from 1 (absolutely disagree) to 5 (absolutely agree). The confirmatory factor analysis demonstrated an excellent fit for the three-factor model, $\chi^2(24)=24.48$, $p=0.44$, CFI=0.998, RMSEA=0.014, SRMR=0.058. The adapted items used in this study are provided in Appendix A as the ChatGPT-Specific Epistemic Beliefs Questionnaire (ChatGPT-EBQ).

Epistemic beliefs about expert-written and ChatGPT-generated articles

After reading the expert-written and ChatGPT-written passages on the effects of vitamin C on the common cold, participants assessed their epistemic beliefs about the content in each text using four items inspired by Mason et al. (2010), each representing a key dimension of epistemic beliefs. The source of knowledge was assessed with the item 'I trusted the author of this text'. The certainty of knowledge was measured with the item 'This text provided factual information that is universally valid'. The structure of knowledge was evaluated with the item 'This text provided complex information'. Finally, the justification for knowing was assessed with the item, 'There was no need to verify the information in this text'. Participants rated each item on a Likert scale ranging from 1 (absolutely disagree) to 5 (absolutely agree). The reliability of the items was $\omega=0.74$ for the expert-written article and $\omega=0.68$ for the ChatGPT-generated article.

Writing task

After reading the expert-written and ChatGPT-generated articles, the participants were instructed to write a short article for the school newspaper. The instructions were as follows:

> Imagine that you are writing a newspaper article for the fall edition of the school newspaper. It is the season of colds, and your task is to write a short article (maximum 2 paragraphs) about the effects of vitamin C in the prevention and treatment of colds. When writing, please use all relevant information from the two articles provided.

The participants' texts were analysed sentence by three trained experimenters. Each sentence was classified as containing (1) correct information sourced from the expert-written article, (2) correct information sourced from the ChatGPT-generated article or (3) incorrect information sourced from the ChatGPT-generated article. The rubric categorising information as correct or incorrect from both texts was developed prior to the coding process. The overall interrater agreement was ICC(3,3)=0.92, indicating excellent consistency among raters.

Analytical procedure

Statistical analyses were conducted using IBM SPSS 29. Dependent variables were the amount of (1) correct information sourced from the expert-written article, (2) correct information sourced from the ChatGPT-generated article and (3) incorrect information sourced from the ChatGPT-generated article.

One-way ANCOVAs were conducted for each dependent variable to examine differences between control and experimental conditions, with (1) prior experience with ChatGPT and (2) prior knowledge about vitamin C included as covariates. Partial eta squared (η_p^2) was reported as a measure of the effect size. Post hoc comparisons were adjusted using the Holm-Bonferroni correction to account for multiple tests.

Hierarchical regression analyses were conducted to evaluate the predictors of information inclusion. $Model_0$ included the experimental condition (independent vs. ChatGPT-assisted writing) as a baseline predictor. $Model_1$ added metacognitive bias and $Model_2$ included epistemic beliefs. The change in R^2 between models was used to assess the unique variance explained by metacognitive bias and epistemic beliefs, respectively.

RESULTS

The Results section presents the findings from the analyses testing the differences between articles written in the control condition (independent writing) and the experimental condition (ChatGPT-assisted writing). One-way ANCOVAs address the hypotheses concerning the inclusion of correct and incorrect information sourced from the expert-written and ChatGPT-generated articles. Subsequently, the Results section reports the hierarchical regression analyses, examining the predictive value of metacognitive bias ($Model_1$) and epistemic beliefs ($Model_2$) on the inclusion of correct and incorrect information from both sources. Descriptive statistics for both groups are presented in Table 2, while linear correlations among variables are provided in Table A1 in the Appendix.

Information in student articles: Comparison of ChatGPT-assisted and independent writing

One-way ANCOVAs were conducted to examine differences between independent and ChatGPT-assisted writing in the inclusion of correct and incorrect information. Prior experience with ChatGPT and prior knowledge about vitamin C were included as covariates.

TABLE 2 Descriptive statistics.

	Control (no ChatGPT)				Experimental (with ChatGPT)			
Variable	*M*	*SD*	MIN	MAX	*M*	*SD*	MIN	MAX
Assessed before experiment								
Metacognitive accuracy (bias index)	0.28	0.43	−0.56	1.00	0.19	0.38	−0.33	0.96
Epistemic Beliefs about ChatGPT								
Certainty of knowledge	2.15	0.68	1.00	3.67	2.25	0.89	1.00	4.67
Structure of knowledge	2.76	0.77	1.00	4.33	2.72	0.94	1.00	5.00
Justification for knowing	4.16	0.71	2.33	5.00	4.19	0.94	1.00	5.00
Epistemic beliefs (task specific)								
Beliefs about expert article	3.80	0.66	1.75	5.00	3.79	0.78	1.25	5.00
Beliefs about ChatGPT article	2.85	0.68	1.25	4.00	2.71	0.69	1.00	4.50
Assessed after experiment								
Amount of information included in student articles								
Correct information (from expert article)	3.99	1.89	0.00	8.00	5.47	2.69	1.00	11.00
Correct information (from ChatGPT article)	1.28	1.02	0.00	4.00	2.64	1.76	0.00	7.00
Incorrect information (from ChatGPT article)	0.87	1.01	0.00	4.00	1.07	1.00	0.00	4.00
Word count of student articles	149.49	54.54	55.00	267.00	190.74	44.91	106.00	311.00

For the inclusion of correct information sourced from the expert-written article, the participants working with ChatGPT included moderately more correct information, $F(1,93)=11.43$, $p<0.001$, $\eta_p^2=0.11$. For correct information sourced from the ChatGPT-generated article, participants working with ChatGPT included significantly more correct information, $F(1,93)=20.41$, $p<0.001$, with a large effect size, $\eta_p^2=0.18$. However, no significant differences were found between the groups for the inclusion of incorrect information sourced from the ChatGPT-generated article, $F(1,93)=0.76$, $p=0.38$, $\eta_p^2=0.01$. The results are visualised in Figure 1.

Overall, these results support H1a and H1b, showing that ChatGPT-assisted writing led to greater inclusion of correct information from both expert-written and ChatGPT-generated articles. However, H1c was not supported, as ChatGPT-assisted writing did not result in a higher inclusion of incorrect information from the ChatGPT-generated article.

Moreover, the student articles written in the experimental condition were longer, $F(1,96)=16.77$, $p<0.001$, with a large effect size, $\eta_p^2=0.15$. The length of the articles was strongly associated with the inclusion of correct expert information ($r=0.59$, $p<0.001$) and moderately associated with the correct ChatGPT-generated information ($r=0.28$, $p=0.005$), but not with the inclusion of incorrect ChatGPT-generated information ($r=0.03$, $p=0.76$).

Predictors of inclusion of correct information from an expert-written article

To test the predictors of correct information inclusion from the expert-written article, hierarchical regression analyses reported in Table 3 were conducted in three steps. In Model$_0$, the experimental condition (independent vs. ChatGPT-assisted writing) moderately predicted the inclusion of correct information, $\beta=0.37$, $p=0.002$, explaining $R^2=9\%$ of the variance.

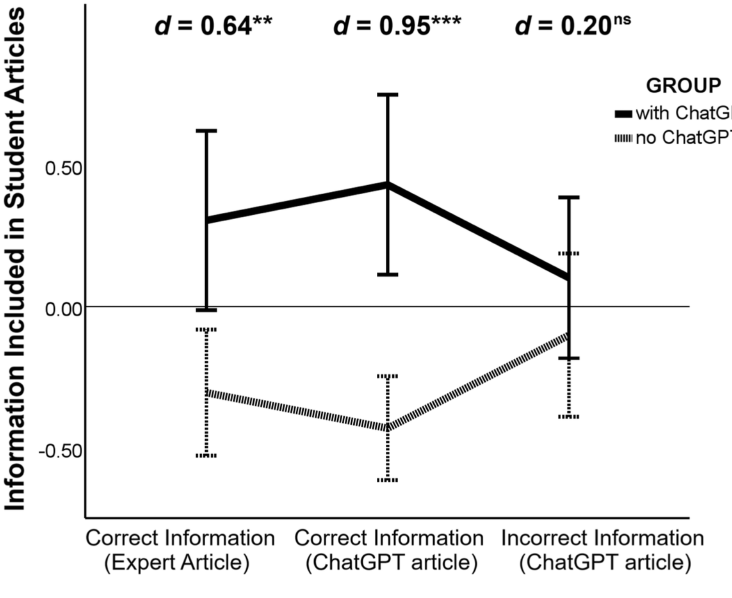

FIGURE 1 Amount of correct and incorrect information sourced from a medical article and a ChatGPT-generated article across control and experimental conditions. The amount of information is standardised to *z*-scores. The raw scores are reported in Table 2. *d* represents Cohen's *d*. The error bars represent 95% confidence intervals. nsnon-significant, $^{**}p<0.01$; $^{***}p<0.001$.

TABLE 3 Variables predicting inclusion of correct and incorrect information in student articles.

	Correct information (expert article)		Correct information (ChatGPT article)		Incorrect information (ChatGPT article)	
	b	**β**	**b**	**β**	**b**	**β**
Model$_0$: CHATGPT						
Experimental condition	1.48**		1.37***		0.20	
R^2	**9%**		**19%**		**1%**	
$F(1, 96)$	9.95**		22.21***		1.02	
Model$_1$: CHATGPT + META						
Experimental condition	1.30**		1.38***		0.28	
Bias index	−2.02***	−0.34***	0.08	0.02	0.78**	0.31**
R^2	**21%**		**19%**		**11%**	
$F(2, 95)$	12.24***		11.02***		5.72**	
ΔR^2	**12%**		**0%**		**10%**	
$\Delta F(1, 95)$	13.26***		0.05		10.32**	
Model$_2$: CHATGPT + META + EPI						
Experimental condition	1.29**		1.43***		0.28	
Bias index	−2.05***	−0.34***	−0.18	−0.05	0.74**	0.30**
Certainty of knowledge	0.08	0.03	−0.21	−0.10	0.12	0.10
Structure of knowledge	−0.64*	−0.23*	−0.03	−0.02	0.04	0.04
Justification for knowing	−0.18	−0.06	0.17	0.09	0.31**	0.26**
Beliefs about expert article	0.36	0.11	−0.30	−0.14	−0.30*	−0.22*
Beliefs about ChatGPT article	0.18	0.03	0.49	0.21*	0.21	0.14
R^2	**26%**		**24%**		**23%**	
$F(7, 90)$	4.49***		4.09***		3.82***	
ΔR^2	**5%**		**5%**		**12%**	
$\Delta F(5, 90)$	1.31		1.27		2.84*	

Note: The bold values are effect sizes (R2) or changes in effect sizes (ΔR2). The significance is provided with the F values and ΔF values, which is the correct way.

*$p<0.05$; **$p<0.01$; ***$p<0.001$.

In Model$_1$, lower metacognitive bias significantly predicted higher inclusion of correct information from the expert-written article, $\beta = -0.34$, $p < 0.001$, increasing the explained variance by $\Delta R^2 = 12\%$.

In Model$_2$, only stronger beliefs about the structure of knowledge generated by ChatGPT were associated with a lower inclusion of correct information from the expert-written article ($\beta = -0.23$, $p = 0.049$). The inclusion of epistemic beliefs explained an additional $\Delta R^2 = 5\%$ of the variance. Interactions between predictor variables were tested but did not reach significance (all $ps \geq 0.75$).

To address RQ1 regarding metacognitive bias, the analysis revealed that a lower metacognitive bias was associated with a higher inclusion of correct information from the expert-written article. For RQ2, only the belief that ChatGPT generates highly complex information was significantly associated with a lower inclusion of correct information from the expert-written article.

Predictors of inclusion of correct information from a ChatGPT-generated article

In Model_0, the experimental condition acted as a strong predictor of the inclusion of correct information, $\beta = 0.45$, $p < 0.001$, explaining $R^2 = 19\%$ of the variance.

In Model_1, metacognitive bias did not significantly predict the inclusion of correct information from the ChatGPT-generated article, $\beta = 0.02$, $p = 0.84$, and the addition of this variable did not increase the explained variance ($\Delta R^2 = 0\%$).

In Model_2, only stronger text-specific beliefs about the ChatGPT-generated article significantly predicted higher inclusion of correct information, $\beta = 0.21$, $p = 0.048$. Epistemic beliefs explained an additional $\Delta R^2 = 5\%$ of the variance. Interactions between predictor variables were tested, but did not reach significance (all $p\text{s} \geq 0.95$).

To address RQ1, metacognitive bias did not significantly predict the inclusion of correct information from the ChatGPT-generated article. For RQ2, stronger text-specific beliefs about ChatGPT-generated content were significantly associated with a higher inclusion of correct information from the ChatGPT-generated article.

Predictors of inclusion of incorrect information from a ChatGPT-generated article

In Model_0, the experimental condition did not significantly predict the inclusion of incorrect information ($\beta = 0.09$, $p = 0.32$), explaining only $R^2 = 1\%$ of the variance.

In Model_1, metacognitive bias was added to the model and emerged as a significant predictor. Higher metacognitive bias (ie, overconfidence) was associated with a greater inclusion of incorrect information, $\beta = 0.31$, $p = 0.002$. Metacognitive bias increased the explained variance by $\Delta R^2 = 10\%$.

In Model_2, stronger beliefs about the expert-written article significantly predicted lower inclusion of incorrect information, $\beta = -0.22$, $p = 0.028$. Interestingly, a higher need to justify the ChatGPT-generated content significantly predicted a higher inclusion of incorrect information ($\beta = 0.26$, $p = 0.010$). The inclusion of epistemic beliefs explained an additional $\Delta R^2 = 12\%$ of the variance. Interactions between predictor variables were tested but did not reach significance (all $p\text{s} \geq 0.50$).

To address RQ1, a lower metacognitive bias was associated with a lower inclusion of incorrect information. For RQ2, stronger beliefs about expert content were linked to a lower inclusion of incorrect information from the ChatGPT-generated article. Interestingly, a higher need to justify ChatGPT-generated content predicted a higher inclusion of incorrect information.

DISCUSSION

As generative AI tools like ChatGPT become more integrated into everyday tasks, it is essential to understand how users engage with and assess the content these tools produce. The purpose of this study was to examine how university students integrate correct and incorrect information from expert-written and ChatGPT-generated articles when writing under two conditions: independently and with the assistance of ChatGPT. Additionally, the study explored how metacognitive bias and epistemic beliefs predicted the inclusion of correct and incorrect information.

Integrating correct expert knowledge

The results showed that the students who wrote with the assistance of ChatGPT included significantly more correct expert knowledge in their articles compared to those who wrote independently ($R^2=9\%$). Furthermore, the texts written with ChatGPT assistance were significantly longer ($d=0.83$). This aligns with prior research highlighting ChatGPT's utility in providing accurate and reliable content for complex tasks (Sabri et al., 2025), resulting in more elaborated and useful outcomes (Noy & Zhang, 2023; Urban et al., 2024). However, the results also revealed that students' metacognitive accuracy played an important role in optimising this integration. Lower metacognitive bias was associated with a higher inclusion of correct expert information, even beyond the baseline improvements offered by ChatGPT assistance ($\Delta R^2=12\%$). These findings suggest that while ChatGPT can enhance performance, students' ability to accurately assess their own thinking remains crucial to effectively integrate information from human-generated sources not included in the training dataset of generative AI tools.

Integrating correct ChatGPT-generated knowledge

Our findings further showed that students working with ChatGPT integrated its correct content more often than those who worked independently ($R^2=19\%$). Moreover, metacognitive bias was not significantly associated with the inclusion of the correct information from ChatGPT, suggesting that integrating the correct ChatGPT-generated knowledge is independent of the users' ability to critically evaluate or calibrate their confidence. This aligns with a recent study by Goh et al. (2024), which highlighted the exceptional accuracy of generative AI in diagnosing clinical cases. In their study, generative AI consistently provided correct answers, but low trust among medical practitioners led them to reject these accurate solutions in favour of less reliable human judgments. Similarly, in our study, the only significant predictor of integrating correct ChatGPT-generated knowledge was trust in the tool, as reflected in stronger text-specific epistemic beliefs about the ChatGPT-generated article.

Integrating incorrect ChatGPT-generated knowledge

As discussed above, ChatGPT excels in delivering correct information when it 'knows', making trust a critical factor in effective use of its capabilities. However, the challenge emerges when ChatGPT generates hallucinations or partially incorrect information (Chelli et al., 2024; Ferrario et al., 2024; Huang et al., 2024). In such cases, the same trust that facilitates the integration of correct information can lead to the acceptance of errors if users fail to critically evaluate the content. The present study showed that ChatGPT assistance neither significantly helped nor harmed in reducing the inclusion of incorrect information in students' writings. However, the results indicate that a higher metacognitive bias was moderately associated with integrating inaccuracies ($\Delta R^2=10\%$). This finding suggests that while ChatGPT assistance was not disproportionately misleading, users' ability to critically evaluate information remains crucial for minimising errors (Mason et al., 2010).

A more nuanced pattern emerged regarding epistemic beliefs. Task-specific epistemic beliefs behaved as expected: lower trust in the expert-written article and higher trust in the ChatGPT-generated article were associated with greater inclusion of incorrect information. Interestingly, however, the relationship between the justification for knowing and the inclusion of incorrect information was counterintuitive. Participants who reported a higher need to evaluate ChatGPT-generated knowledge were more likely to include inaccuracies.

Although unexpected, the findings showed a similar pattern to that reported in Strømsø and Bråten (2010). Their study examined how epistemic beliefs predicted students' problem-solving, help-seeking and self-regulatory strategies when using the Internet for learning. Students with strong beliefs about justification for knowing negatively predicted self-reports of self-regulation. This finding may reflect the cognitive demands of critical evaluation, as students who prioritise reasoning must allocate significant cognitive resources to assess the reliability and consistency of information (Anmarkrud et al., 2019). Furthermore, the negative relationship might also reflect a potential trade-off: students deeply engaged in evaluating knowledge either perceive their self-regulatory strategies as less effective (Boekaerts & Corno, 2005; Zimmerman & Martinez-Pons, 1992), or underestimate them (Seban et al., 2024), especially when dealing with conflicting information (Maier & Richter, 2014; Stadtler & Bromme, 2007).

Implications

The lack of significant interactions between predictors in this study suggests that observed patterns are consistent across both independent writing and ChatGPT-assisted groups. This finding indicates that metacognitive bias and epistemic beliefs are equally relevant for information integration, regardless of whether students work independently or with the aid of generative AI tools.

The first key implication is the need to develop an understanding of the generative AI tools' strengths and weaknesses. For example, as Goh et al. (2024) found in the context of medical practitioners, trust in ChatGPT is essential when the tool 'knows' and delivers accurate information. In such cases, distrust can hinder effective outcomes, as the tool often performs remarkably well on its own. However, users must recognise the limitations of generative AI, particularly its tendency to hallucinate or produce incorrect information. When working with content that goes beyond ChatGPT's capabilities or dealing with potentially erroneous outputs, metacognitive skills become crucial. In these scenarios, the ability to critically evaluate and verify information is essential to avoid integrating inaccuracies (Barzilai & Ka'adan, 2017).

Furthermore, as argued by Lang et al. (2021), domain-specific epistemic beliefs about the uncertainty of knowledge provide a stable foundation for reasoning, while task-specific epistemic beliefs translate these general principles into actionable evaluations. Students with stronger epistemic beliefs about uncertainty performed better because they were more inclined to consider multiple perspectives and critically assess conflicting claims. In the context of our findings, this dual operation highlights the importance of both domain-specific and task-specific beliefs when working with generative AI tools. However, in line with Strømsø and Bråten (2010), the present study suggests that domain-specific beliefs must also be calibrated to discern when evaluation is redundant and when it is, in fact, necessary.

Limitations

While this study addressed the role of metacognitive bias and epistemic beliefs in integrating information from expert-written and ChatGPT-generated articles in university students, several limitations should be considered. First, the study does not account for developmental differences in epistemic beliefs and metacognitive processes. Prior research suggests that these constructs evolve over time and are influenced by cultural context and educational progress (Hofer, 2008; Hofer & Pintrich, 1997). As our sample consisted of university students, future studies should include individuals at different educational stages. Second, in the present study, epistemic beliefs were treated as static, yet they may be influenced by situational factors, such as task demands and repeated interactions with information sources (Hofer & Sinatra, 2010). Finally, the study relied on self-reports and separate-task measures of metacognitive bias, which may not fully reflect the dynamic nature of these constructs.

Future directions

Fostering informed trust—combined with strategies for evaluating potentially misleading outputs—remains essential for maximising ChatGPT's utility while mitigating its limitations. This underscores the need for individuals to approach AI-generated content with a balance of trust and scepticism, applying evaluation skills to discern reliable information. Educators play a pivotal role in this process, as fostering epistemic awareness in learners is crucial to mitigating the risks of misinformation (Rapp & Withall, 2024).

Educators should design argumentation tasks that explicitly require source evaluation, justification of expert-written and AI-generated knowledge, and deep elaboration to activate epistemic beliefs (Bromme et al., 2010; Greene et al., 2018). Drawing on insights from the COPES model (Winne & Hadwin, 1998), epistemic beliefs can serve as internal standards that guide learners in adjusting their approaches to task demands, enhancing metacognitive accuracy and adaptability (Bromme et al., 2010). Interventions that encourage reflection on the nature of knowledge and strategic responses to task complexity can further empower learners to regulate their reasoning (Barzilai & Ka'adan, 2017; Bromme et al., 2010) and better engage with AI-generated information. Moreover, metacognitive interventions focusing on uncertainty monitoring, evaluation of information, and regulation of reasoning processes alongside epistemic awareness can better equip students to understand the intricacies of AI-generated information (Lang et al., 2021; Salovich & Rapp, 2021). These interventions should be structured to include tasks that encourage students to verbalise their thought processes and reflect on their epistemic beliefs during and after task performance, such as through cued retrospective verbal reports or guided think-aloud activities. This approach allows learners to recognise and address uncertainty of knowledge, develop strategies for evaluating controversial claims, and regulate their reasoning processes more effectively (Lang et al., 2021). For instance, strategic scaffolds, which provided guidance on how to evaluate and integrate sources, improved integration performance, while metastrategic scaffolds further enhanced students' understanding of when, why, and how to apply epistemic strategies (Barzilai & Ka'adan, 2017). Moreover, educators can design tasks that incorporate conflicting knowledge claims, prompting learners to critically analyse the justifications and evidence behind various perspectives, thus enhancing their epistemic cognition and metacognitive accuracy (Bråten et al., 2016).

Finally, future research should employ mixed-method approaches to investigate the cognitive processes underlying information integration. Qualitative interviews could explore students' experiences and strategies when using generative AI: think-aloud protocols could capture real-time decision-making (similarly to Strømsø et al., 2013) during human-AI collaboration, and eye-tracking combined with process-mining could offer detailed insights into attention allocation and information selection.

CONCLUSION

Generative AI enters a society where scientific truth is contested, with an increasing number of people holding unscientific beliefs and exhibiting strong resistance to evidence-based reasoning (Hornsey et al., 2018; Hornsey & Fielding, 2017). This context amplifies the need for fostering epistemic beliefs that support critical engagement with complex information. By equipping individuals with the skills to discern reliable from unreliable information—both human-generated and AI-generated—educators can address these challenges. Research on epistemic beliefs and metacognition offers valuable insights for designing interventions that help individuals combat misinformation and make informed decisions grounded in evidence (Hartman et al., 2017; Wilson, 2018). Strengthening these competencies is critical not only for working effectively with AI, but also for promoting a broader culture of critical thinking and scientific literacy in an era of rapidly advancing technology.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The research was conducted in conformity with APA's ethical code. Ethical approval was obtained from an ethical committee of the Institute of Psychology, The Czech Academy of Sciences, under the number PSU-530/Brno/2024. All participants provided their own written consent.

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APPENDIX A

ITEMS OF CHATGPT-SPECIFIC EPISTEMIC BELIEFS QUESTIONNAIRE (CHATGPT-EBQ)

Certainty and source of knowledge

1. The truth about almost every issue raised in my classes can be found using ChatGPT.
2. ChatGPT can provide me with most of the knowledge I need to succeed in my courses.
3. Most of what is true in my field of study is available through ChatGPT.

Structure of knowledge

1. The strength of ChatGPT lies in the vast amount of detailed information it provides about what I am studying.
2. On ChatGPT, the richness of detail about what I am studying is most prominent.
3. The most important aspect of ChatGPT is that it provides so many specific facts about what I am studying in my classes.

Justification for knowing

1. To find out whether the course-related knowledge that I get from ChatGPT is trustworthy, I try to compare it with knowledge from multiple sources.
2. I evaluate whether the course-related knowledge that I get from ChatGPT seems logical.
3. I evaluate course-related knowledge claims that I encounter on ChatGPT by checking other knowledge sources about the same topic.

Adapted from Strømsø and Bråten (2010).

TABLE A1 Linear correlations between variables.

	Variable	1	2	3	4	5	6	7	8
1.	Metacognitive accuracy (bias index)	—							
Epistemic beliefs (person level)									
2.	Certainty of knowledge	−0.10	—						
3.	Structure of knowledge	−0.02	0.53***	—					
4.	Justification for knowing	−0.17†	−0.08	0.04	—				
Epistemic beliefs (task specific)									
5.	Beliefs about medical article	−0.16	−0.06	0.00	0.25*	—			
6.	Beliefs about ChatGPT article	0.25*	0.30**	0.39***	−0.13	−0.01	—		
Amount of information included in student articles									
7.	Correct information (from medical article)	−0.37***	−0.03	−0.20*	0.01	0.14	−0.15	—	
8.	Correct information (from ChatGPT article)	−0.03	−0.01	0.01	0.05	−0.11	0.11	−0.04	—
9.	Incorrect Information (from ChatGPT Article)	0.30**	0.13	0.14	0.13	−0.21*	0.22**	−0.34***	0.33***

†$p < 0.10$;
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.